

CSE 5095: Types and Programming Languages

Rule Induction and Syntax

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September 3, 2026

1 Rule Induction

Below are the proofs done in class, written out formally as you would write them on a homework.

First, here are the rules for our “is XML” judgment.

$$\frac{}{\text{"" is XML}} \text{ (X-1)} \quad \frac{}{\langle T \rangle \text{ is XML}} \text{ (X-2)} \quad \frac{X \text{ is XML}}{\langle T \rangle X \langle /T \rangle \text{ is XML}} \text{ (X-3)} \quad \frac{X \text{ is XML} \quad Y \text{ is XML}}{X Y \text{ is XML}} \text{ (X-4)}$$

Let $OpenAngle(X)$ be the number of open angle brackets in a string X and $CloseAngle(X)$ be the number of close angle brackets in X .

Theorem 1. *If X is XML, then $OpenAngle(X) = CloseAngle(X)$.*

Proof. By induction on the derivation of X is XML.

- Case X-1. Then $X = \text{""}$ and $OpenAngle(X) = CloseAngle(X) = 0$.
- Case X-2. Then $X = \langle T \rangle$ and $OpenAngle(X) = CloseAngle(X) = 1$.
- Case X-3. Then $X = \langle T \rangle Y \langle /T \rangle$ and Y is XML. We have $OpenAngle(X) = 2 + OpenAngle(Y)$ and $CloseAngle(X) = 2 + CloseAngle(Y)$. By induction, $OpenAngle(Y) = CloseAngle(Y)$, so $2 + OpenAngle(Y) = 2 + CloseAngle(Y)$.
- Case X-4. Then $X = Y Z$ and Y is XML and Z is XML. We have $OpenAngle(X) = OpenAngle(Y) + OpenAngle(Z)$ and $CloseAngle(X) = CloseAngle(Y) + CloseAngle(Z)$. By induction, $OpenAngle(Y) = CloseAngle(Y)$ and $OpenAngle(Z) = CloseAngle(Z)$, so $OpenAngle(X) = CloseAngle(X)$.

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Now here are the rules for natural numbers (both constructing a natural number and the greater-than judgment)

$$\frac{}{0 \text{ is a natural number}} \text{ (ZERO)} \quad \frac{n \text{ is a natural number}}{n + 1 \text{ is a natural number}} \text{ (SUCC)} \quad \frac{n \text{ is a natural number}}{n \geq n} \text{ (GE-NAT)}$$
$$\frac{m \text{ is a natural number} \quad m \geq n}{m + 1 \geq n} \text{ (GE-SUCC)}$$

Theorem 2. *If n is a natural number then $n \geq 0$.*

Proof. By induction on the derivation of n is a natural number.

- Rule ZERO. Then $n = 0$. By GE-NAT, we have $n \geq 0$.
- Rule SUCC. Then $n = m + 1$ and m is a natural number. By induction, $m \geq 0$. By GE-SUCC, we have $m + 1 \geq 0$.

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